

Zero to LLM

Build a language model from scratch. On your own data.
For less than a dollar.

We will train one live, in this room, in under five minutes.



Neil Celik

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THINGS I'VE ACTUALLY SHIPPED

Co-founded **UrbanStat** — machine learning for property insurance. We built **holistic risk scores** for individual buildings, on top of our own peril models for **wildfire, landslide, earthquake and flood**. Now used by P&C carriers across the US, Turkey and Portugal to decide what they will and won't underwrite. Earlier, built forecasting and optimisation software for **cash-in-transit operations** — routing and cash levels across ATM and branch networks — still running inside multinational banks.

SOME COMPANIES I WORKED WITH

BNP Paribas Allianz ING Ageas Sampo Japan Safety Insurance Gulf Insurance Group Vienna Insurance Group

THE FINANCE AND OPERATIONS SIDE

Chief of Staff to the CEO at **Logiwa** (Series B SaaS): owned FP&A end to end, negotiated an **eight-figure venture debt facility**, passed GAAP-compliant audits, and stood up **SOC 2 Type II from scratch in under three months**. Acting CFO along the way.

SOME COMPANIES I WORKED WITH

Flexport Ware2Go Radial

THE RESEARCH BEHIND TODAY

Booth MBA — Finance, Analytic Finance, Applied AI. My final-year research built a novel **asset pricing factor** out of proprietary language models, measuring how a firm's profitability compares against **text-based peer groups**. Mentored by **Eugene Fama** — Nobel laureate, father of the **efficient-market hypothesis** and co-creator of the **Fama–French factor models** that are the industry standard for telling genuine skill from risk in disguise. It's his five-factor model that Part 2 builds on.

From a stack of SEC filings to a model trained on nothing else.

Five parts. You do not need to write a line of code — watch, interrupt me, and take the repo home. The one thing worth knowing up front: the model starts training in Part 02 and we talk over it, so by Part 04 it is alive.

PART 01

What a language model actually is

Stripped of the mystique, it is one sentence. I'll prove it rather than assert it.

PART 02

Why on earth would you train your own?

The research this all came out of — and the cases where you honestly shouldn't bother.

TRAINING STARTS HERE

PART 03

How did this get 900× cheaper?

The same GPT-2-class model: \$43,000 in 2019, about \$48 today. Nothing in the mathematics changed.

PART 04

It's alive

We open the model that has been training while we talked, and take out what it learned.

LIVE MODEL

PART 05

Where would this fit in a deal?

Five places across your lifecycle where something like this might earn its keep.

\$43,000

GPT-2 · 2019 · A RESEARCH LAB



\$48

SAME CAPABILITY · TODAY · RENTED BY THE HOUR

That is the same model quality, seven years apart — roughly **900× cheaper**.
Nothing about the mathematics changed. Seven years of compounding improvements across the whole stack did.

The model we train together in this room costs about **\$0.20**.

— PART 01

What a language model actually is

It's autocomplete — that had to learn what the words mean.

You hand it some text; it predicts what comes next. Then it does it again. And again. That really is the whole machine.

But “just autocomplete” undersells it badly. To guess the next word well, it has no choice but to work out grammar, context, tone, and what the words actually *refer to*. You cannot finish this sentence without knowing what a cat is, and what sort of thing a cat sits on.

The cat sat on the



The model scores *every* word it knows, then picks one. Usually the top. Sometimes it rolls the dice a little — that's what “temperature” means.

Though if you've spent time in Istanbul, as I have, you'd have put rather more weight on “sidewalk” — and you would not be wrong. **The right answer depends on which cats you've met.** So does the model's.

Four steps, on repeat

- 01 **Tokens.** Chop the text into chunks. "The cat sat" → [The][cat][sat]. Ours invents a private vocabulary of **4,096** chunks. **GPT-4** used 100,277; **GPT-4o** and its successors use **200,019**. *Anthropic doesn't publish Claude's — which tells you it's considered worth keeping quiet.*
- 02 **Vectors.** Turn each chunk into a long list of numbers, and place it in space. Words used alike land near each other — and, remarkably, **the gaps between them carry meaning too**. The step from *man* to *woman* is the same step as from *king* to *queen*. Nobody built that. It fell out of the guessing.
- 03 **Attention.** Every chunk looks back at the chunks before it, decides which matter, and **rewrites its own vector to fold them in**. For "the cat sat on the ___" the blank pulls hard on *cat* and *sat*. It's the *vectors* that change here — not the model's weights, which were fixed when it trained. This mix repeats through dozens of layers.
- 04 **Probability.** Score all 4,096 chunks. Pick one. Stick it on the end of the sentence. **Then go back to step 1 and do the whole thing again.**

the same step, in the same direction



Nobody teaches it anything

Take a pile of text. Hide the next word. Ask the model to guess. It guesses wrong. Calculus works out which of its millions of internal numbers caused the mistake, and nudges each one slightly in a better direction.

Repeat. Billions of times.

Grammar, style, world knowledge, the exact shape of a risk-factor disclosure — none of it is programmed. It all falls out of **guess, get corrected, repeat.**

When training stops, the numbers freeze. Run the frozen model in a loop and you get essays, code, or a very convincing 10-K.



Loss = how wrong the guess was. Around **8**, the model is guessing at random. Below **2**, it has learned how a filing sounds. You'll watch it fall live — the training itself takes about twenty seconds.

The text it writes is the boring part

To predict the next word in a document, the model is *forced* to build an internal picture of what that document is about. Not a summary — a list of 256 numbers. A fingerprint. That fingerprint is called an **embedding**.

For a model this small, the text it writes is a party trick — **the embeddings are the real product**. Every result in my research came out of these numbers, not out of a single word the model ever wrote.

SIMILARITY

Which of these 9,438 filings read like this one? Not by keyword — by meaning.

CLUSTERING

What themes are in this corpus that nobody has named yet?

DRIFT

How has this company's language changed over twenty years — and when did it turn?

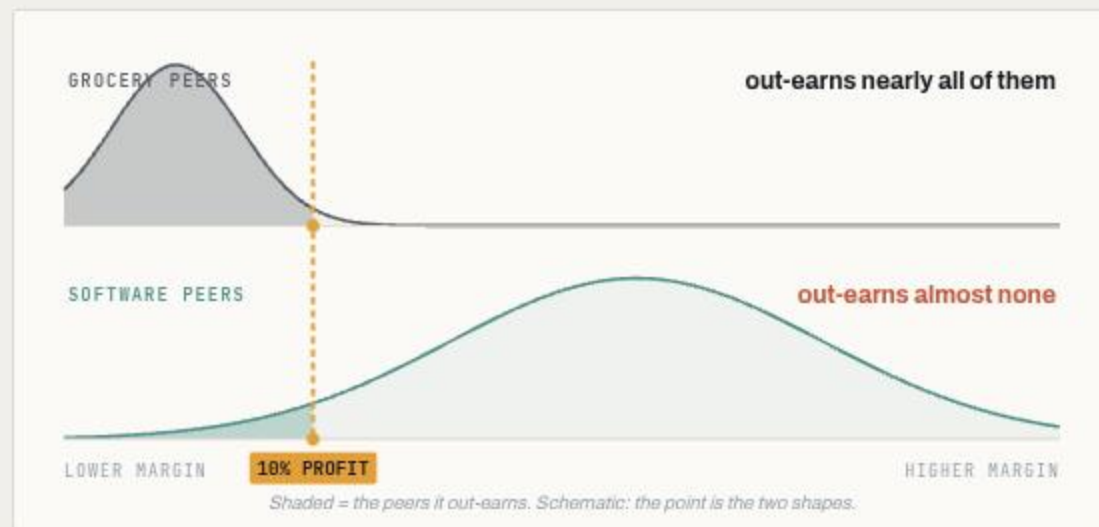
— PART 02

Why on earth would you train your own?

► START TRAINING NOW • IT RUNS WHILE I TALK

Usually under five minutes, but it can take fifteen. It'll be done well before Part 4 — and if it isn't, there's a model already saved on the server we can talk to instead.

Do investors read 10% profit the same way at a grocer and at a software firm?



Same number. **Opposite verdict.** The question isn't whether *you* know that — it's whether **the market** knows it, and prices it.

EFFICIENT MARKETS SAY

A dollar of profit is a dollar. The market read the same filings you did, so any peer comparison is *already in the price*. Nothing left to find.

BEHAVIOUR SAYS

Investors anchor on whichever reference group is handed to them. **Open a Bloomberg terminal, look at the “comparable companies” panel.** Somebody else chose that list. We don't know how, from what, or when it last changed. **Everyone uses it anyway.**

WHERE I THINK IT SITS

In between — and that's testable, not philosophical. Mostly priced; not entirely. Any residual shows up as a factor the five-factor model cannot explain.

Which turns it all on one question: **compared to whom?** The official answer is a code assigned decades ago — SIC — and it is broken two separate ways.

Two problems, one of them fatal

PROBLEM 1 • THE LABEL IS STALE

Amazon is still filed as “mail order”

Roughly 90% of SIC codes **never change** once assigned. Amazon's is 5961 — *mail order / catalog retail*. Amazon also runs the largest cloud business on earth.

And firms are multi-faceted: Amazon is retail *and* cloud *and* advertising *and* logistics *and* healthcare. One static code cannot hold that.

PROBLEM 2 • THE MODEL ALREADY KNOWS THE ENDING

You cannot ask Claude what it didn't know

Ask any frontier model to describe Amazon *as of 2010*. It knows about AWS's rise, Whole Foods, the 2022 crash. **It cannot un-know them.**

That's **look-ahead bias** — you have smuggled the future into a model of the past. Harmless in a demo. Fatal in a backtest that moves money. **No prompt removes it** — it's baked into the weights. (Glasserman & Lin, 2024, document exactly this channel for pre-trained models in finance.)

There was no model on earth I could *prove* hadn't already **seen the future**. So I had to build one that *couldn't* have.

So: one model per year

● ● ● A TIME CAPSULE PER YEAR

```
equityAnalyst2000 ← 1996-1999 only
equityAnalyst2001 ← 1996-2000 only
equityAnalyst2002 ← 1996-2001 only
⋮
equityAnalyst2026 ← 1996-2025 only
```

Thirty encoders, 1996 through 2025. Each one trained from scratch on the 10-Ks and 10-Qs accepted *strictly before* its own year. Same architecture every time — 576 dimensions, 9 attention heads, an 8,000-chunk filing vocabulary. Only the training text moves.

The model that picks Amazon's peers in 2003 **has never read a filing from 2003 or later**. Not because it promised. Because it never saw one.

Nobody ever told it what an industry is



64.3%

OF CHOSEN PEERS LANDED IN THE SAME INDUSTRY DIVISION

It never saw a single SIC code — rows are the official industry, columns the peers *it* chose from language alone. Each row's "Other" is broken out on the right into its top three real industries: it isn't noise.

Where it "gets it wrong," it's arguably right

WATER TRANSPORT

The model's top outside-peer is **Oil & Gas Extraction (15.8%)**.

Because a shipping company's filing is *about* fuel costs, charter rates and commodity exposure. It reads like an energy company. It arguably *is* one.

PIPELINES

Top outside-peers: **Wholesale-Nondurable (17.9%)** and **Petroleum Refining (14.7%)**.

The official code says "Pipelines." The model says "this is the oil business," which is what a human analyst would say.

LOCAL TRANSIT

Top outside-peer: **Health Services (14.2%)**.

Surprising — until you remember how much of local transit is non-emergency medical and paratransit. The risk language gives it away.

A classification system draws hard borders once and defends them for forty years. **The model draws its borders from what the documents actually say — and redraws them every year.**

Watch the model watch Amazon turn into a cloud company



Each column is one year; each cell is the share of Amazon's top-20 model-chosen peers drawn from that industry. Only the three rows that carry the story are shown — the other 35 are near-empty.

A DOT-COM

Amazon's peers are **prepackaged software** — 49% of them in 1999. The market, and the filing, treat it as a tech company. Not a shop.

A STORE

Mail-order retail takes over from 2003, peaking at 43% in 2010. Meanwhile computer services collapses to 5%.

BOTH AT ONCE

Computer services climbs back — 18% in 2013 to 31% by 2018 — and now runs *alongside* retail. The model sees two companies in one.

Amazon's official SIC code did not change once in those 26 years. **The filings changed — and the model heard it.**

4.57% a year that the standard model cannot explain.

Finance has a standard scoreboard for what *should* drive a stock's return. Anything it **can't** explain is called **alpha** — return you were not supposed to be able to earn. I changed exactly one input on that scoreboard: compare a firm's profits to its **text-identified peers** rather than to the whole market. Nothing else.

+4.57%

A YEAR, UNEXPLAINED
T = 5.0 — THE BAR FOR A NEW
FACTOR IS 3

0.63 → 0.88

RETURN PER UNIT OF RISK
THE SAME MONEY, ~40% HARDER-
WORKING

Positive in **every** sub-period — including **2016–2025**, a decade the model was never shown. It didn't get to peek at the answers.

WHAT THIS ACTUALLY SAYS

Everyone already knew profitable firms tend to earn more — that has been priced for years. What nobody was pricing is being profitable **relative to the right peers**. That part was sitting there unclaimed.

AND THE HONEST BOUNDARY

Run the same peer trick on **value** instead of profitability and it falls apart out of sample. The effect is specific, not a free upgrade. **A result that worked everywhere would be a bug, not a triumph.**

Celik, N. (2026), "Profitability, Relative to Peers: Text-Identified Reference Groups and the Cross-Section of Returns." Developed in Research Projects: Finance, taught by Eugene F. Fama, Chicago Booth.

Four reasons to train your own model

01 • SOVEREIGNTY

The data that can't leave



Deal rooms. Working papers. Client files under privilege. A model you trained runs inside your perimeter and never phones home.

02 • VOCABULARY

Fluent in your jargon, not the internet's



It won't *write* better than a frontier model — nothing this small will. It spends every parameter telling *your* documents apart. Worth testing. Not worth assuming.

03 • DEFENSIBILITY

An answer that survives review



"The model said so" doesn't hold up in a review. Own the training set and you can state what went in — and **prove** what it could not have known.

04 • ECONOMICS

Cheap enough to be wrong



Twenty cents buys a toy. The thirty encoders behind that alpha came to a few thousand dollars — **small enough that nobody had to approve it.**

Now let me talk you out of it

TRY THIS FIRST

Write a better prompt

Most "AI projects" turn out to be one good prompt and a spreadsheet. Skipping this because it feels too easy is an expensive kind of pride.

AN AFTERNOON

THEN THIS

Search your own documents

Need facts *out* of documents? That's retrieval, not training. Training a model to memorise your files is the expensive way to build a worse search engine.

A WEEK

THEN THIS

Fine-tune a model that already exists

Start from Llama, Qwen or Mistral. You inherit the world knowledge free and pay only for your own jargon. **Honestly — this is the right answer more often than what we did today.**

A FEW DAYS

ONLY THEN

A NARROW DOOR

Train from scratch

What we did today. It earns its keep in three situations, and no others:

- 1 · the representation *is* the product — similarity, clustering, drift
- 2 · the data genuinely cannot leave the building
- 3 · you must **prove what the model knew, and when**

CHEAPER · FASTER · USUALLY ENOUGH

MORE EXPENSIVE · OCCASIONALLY NECESSARY

— PART 03

**So how did this
get 900× cheaper?**

nanochat



Andrej Karpathy

Founding member of OpenAI. Former Director of AI at Tesla. Released **nanochat** in October 2025 — the from-scratch ChatGPT pipeline, about 8,000 readable lines — then **autoresearch** in early 2026. In May 2026 he joined Anthropic's pre-training team.

“The best ChatGPT that \$100 can buy.”

— the repo's own tagline

GPT-2	2019	• weeks	• a research lab	\$43,000
nanochat speedrun	today	• ~2 hrs	• 8×H100, rented	\$48
		• same, on spot instances		~\$15
What we run here	under 5 min	• 1 GPU		\$0.20

Same capability. Seven years. Nothing in the mathematics changed — better data, better optimisers, better kernels, better hardware, all compounding at once.

AND IT'S GETTING STRANGER — AUTORESEARCH

Point an AI coding agent — the same Claude Code you're watching today — at a training repo, describe what to try in a text file, and it runs the experiments *itself*, keeping only the changes that beat the best. The person who built nanochat now mostly directs agents.



github.com/karpathy/nanochat

Free, MIT-ish, and readable in an evening.

This is the engine under today's demo.

Under five minutes, start to finish

You've watched these tick past in the panel while I was talking about peer groups. Here's what each of them actually did.

● ● ● TRAIN.IPYNB

```
01 rent a GPU — one A100 80GB, by the minute 60s
02 upload the filings 10s
03 install nanochat — the slow part; it is nearly all download ~90s
04 train our own tokenizer — a private 4,096-word vocabulary 30s
05 pretrain the model — 7.3M params, 400 steps 20s
06 generate text from it 15s
07 embed all 9,438 company-years 60s
08 kill the pod — do not skip this one 2s
```

THE CORPUS

46,504 filing sections from 2025–2026 10-Ks — Business, Risk Factors, MD&A and Market Risk. About a gigabyte of the driest prose ever written.

THE MODEL

7.3 million parameters — a rounding error next to GPT-4. It will still learn to write like a lawyer.

STEP 04, QUIETLY

We train the *vocabulary* from scratch too. "**stockholders**" becomes a single token — because in *this* corpus, it earns one.

— PART 04 · LOOK TO YOUR RIGHT

It's alive.

While I was talking, a language model that did not exist half an hour ago finished training in the corner of your screen.

8.3 → 1.7

LOSS · RANDOM GUESSING → FLUENT

< 5 min

START TO FINISH

\$0.20

TOTAL COST

Nobody wrote it a rule. It read 46,504 fillings and worked out how they sound.

It sounds exactly like a filing. Every fact in it is invented.

● ● ● A REAL CONTINUATION FROM THE MODEL WE JUST TRAINED

you > ITEM 1A. RISK FACTORS — The following risks could materially affect our

...it continues:

financial position, results of operations and liquidity. We seek to limit the effects of changes in interest rates and inflation on our cash flows. We do not enter into derivatives or other financial instruments for trading or speculation purposes.

It has the **cadence** of a filing — the hedging, the dread, “materially adversely affect”. Ask it a question and it won't answer; it will just keep writing. **As prose, it's worthless.** But the prose isn't what we came for.

SAY THIS OUT LOUD

Trust **nothing it writes**. At 7.3 million parameters it will state confident financial facts that are entirely made up. **A beautiful, fluent liar.**

AND YET — IT DID LEARN SOMETHING REAL

To guess the next word of Pfizer's risk section, it first had to work out what kind of company Pfizer is. That much is genuinely in there. It simply has no way to say it.

TRY THESE IN THE PANEL

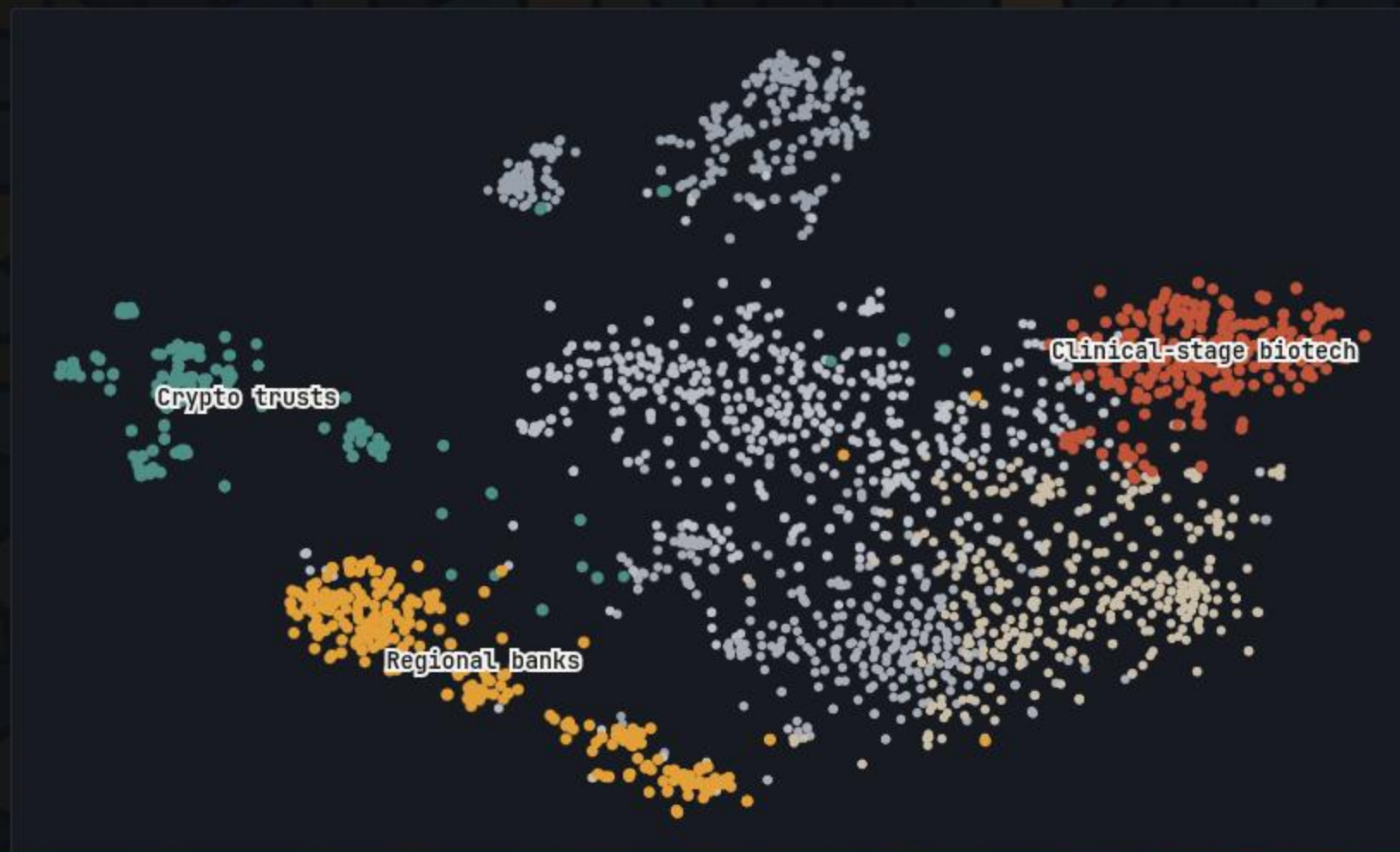
“Our principal sources of revenue are”

“We may be unable to”

“Competition in our industry”

Now we throw the words away

We pushed all 9,438 company-years back through the model and kept only the internal fingerprint of each — 256 numbers per company. Not what it writes. What it *understood*. Then we flattened those onto a page.



NOBODY LABELLED ANY OF THIS

Every dot is one company-year. The model was shown text and nothing else — no industry codes, no sectors, no tickers. **The clumps are its own.**

SQUEEZE IT AND ALMOST NOTHING IS LOST

Throw away 241 of the 256 numbers and you still keep **67%** of everything that separates one company from another. The model gave itself 256 dimensions and only really needed about 15 — **it found a handful of things that genuinely matter.**

(That's about the *shape* of the space — not the number of clusters.)

t-SNE projection of a 2,169-point stratified sample; colours are k-means clusters found across all 9,438. Grey clusters are real too — just harder to name honestly.

Name a company. Get its peers — by meaning.

No keywords, no industry codes, no lists. Just: whose filing reads most like this one? These are real results from the model we trained in this room.

PFE →

AMGN	0.99
GILD	0.99
INDV	0.98
OGN	0.98

Amgen, Gilead, Organon. **All pharma.** The word “pharmaceutical” shows up in 0.1% of the corpus — and never once in Pfizer’s own section.

JPM →

HWBK	0.98
MCO	0.98
CMPS	0.98
FFBC	0.98

A community bank, a ratings agency, a regional bank. **Financials.**

XOM →

NPWR	0.95
IMO	0.95
FMC	0.95
CHX	0.95

Imperial Oil, ChampionX. **Energy.**

DAL →

ALK	0.98
AAL	0.98
JBLU	0.98
ALGT	0.98

Alaska, American, JetBlue, Allegiant. **Four out of four airlines.**

What that just cost

GPU · a few minutes @ \$1.39/hr	\$0.19
Storage & bandwidth	\$0.01
Software licences	\$0.00
Total	\$0.20

Ours is a small model — a toy, and I'll keep saying so. But the collapse is real at every size: **the GPT-2 that cost \$43,000 and a research lab in 2019 is about \$48 today.**

So the interesting question is no longer *“can we afford to?”*

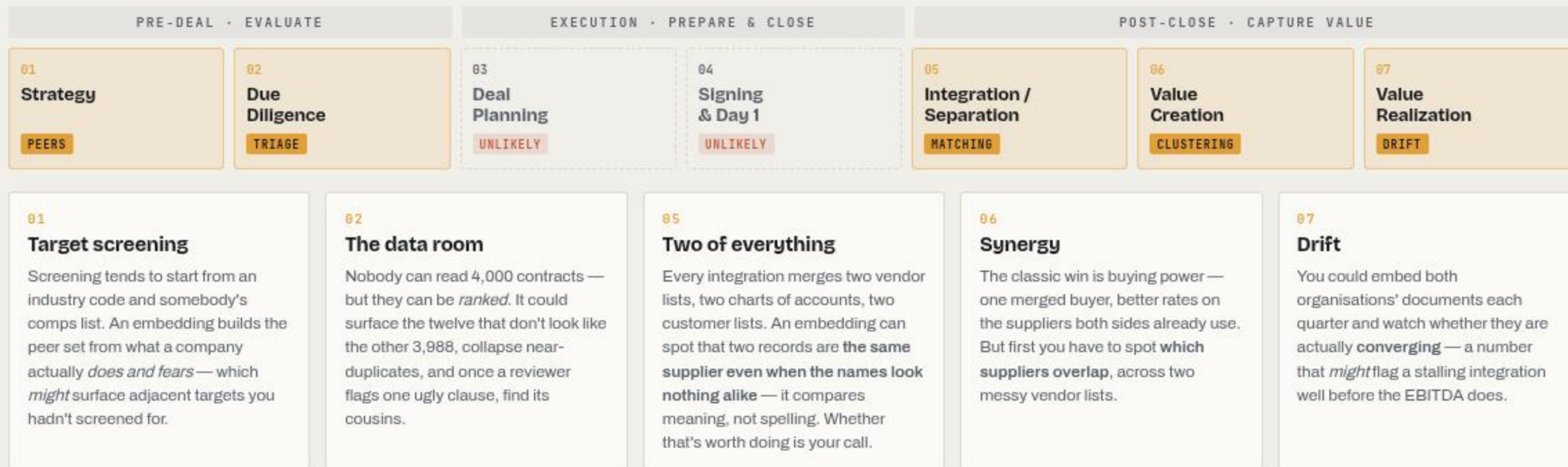
It's “what do we do with it?”

— PART 05

**So where would
this fit in a deal?**

Potential use cases in the deal stages

Everyone in this room has done more deals than I have, so take these as suggestions rather than findings. **Five** places where a model like the one we trained today might earn its keep.



Three things

- 01 **Find the corpus you own that nobody else has.** Not the data you can buy — the data you accumulated by doing the work. *That* is the asset. The model is the cheap part.
- 02 **Ask whether you need an answer or a representation.** Answers come from a prompt or a search. Representations — similarity, clustering, drift, peer sets — are what you train for at this scale.
- 03 **Then just run it.** The version you saw today is a toy — but it's a *twenty-cent* toy, and you do not need a business case for twenty cents. Build the toy, see whether the embeddings separate anything you care about.

THANK YOU

Neil Celik

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The code and the filings are on the next slide, if you ever want them.

— TAKE IT HOME

If you ever want to do this yourself

DO THE WORKSHOP



The code
github.com/ancelik/zero-to-llm



The filings
46,504 SEC sections.
~1 GB, Google Drive.



Code mirror
Same code on Drive,
if GitHub is blocked.

GO DEEPER



nanochat
Karpathy's engine.
Readable in an evening.



"Deep Dive into LLMs"
3.5 hours. The highest-ROI
video on the subject.



Claude Code
Install guide. It'll walk you
through the errors.

You'll want a RunPod account with about **\$10 of credit**. The run itself costs cents.